

Network Forensics

Week 13 Lab: Sysmon Threat Analysis

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Abstract

This report documents a Sysmon-based threat hunt conducted on a Windows 10 host as part of the Week 13 Network Forensics lab. Four Windows Event Log files (EVTX format) were imported into Security Onion and analyzed using the Hunt dashboard. The investigation identified nc.exe (Netcat) as a malicious executable delivered to the system via Google Chrome. Following its download, nc.exe initiated an outbound network connection to an Amazon AWS server on port 80, consistent with command-and-control (C2) communication. Windows Defender (MsMpEng.exe) was found to have modified the file creation timestamp of nc.exe. Additionally, a registry key modification linked to Explorer.EXE was identified under the FileExts autorun path, indicating a persistence mechanism. Taken together, the findings describe a structured attack chain from initial delivery through network callback and persistence establishment.

Introduction

Objective

The objective of this lab is to analyze Sysmon event logs from a Windows 10 host using Security Onion to identify indicators of compromise and reconstruct the attacker activity chain. The investigation addresses five specific questions covering the identity and delivery of a malicious executable, its network behavior, file timestamp modification, and registry-based persistence.

Background

Sysmon (System Monitor) is a Windows system service published as part of the Microsoft Sysinternals suite. It records detailed system activity to the Windows Event Log and provides significantly greater forensic visibility than standard Windows Event Logging alone. The following Sysmon event IDs are relevant to this investigation:

- Event ID 1 (process_creation): Records new process creation with full command line and parent process context.
- Event ID 2 (file_create_time_changed): Detects when a process explicitly modifies a file timestamp, known as timestomping.
- Event ID 3 (network_connection): Logs TCP/UDP connections linked to the originating process via ProcessGUID.
- Event ID 11 (file_create): Records file creation or overwrite events, useful for tracking dropped files.
- Event ID 12/13 (registry_key_create / registry_value_set): Captures registry object creation and value modifications.
- Event ID 15 (file_create_stream_hash): Detects Alternate Data Stream (ADS) creation, which Chrome attaches to all downloaded files as a Zone.Identifier tag.

Security Onion aggregates Sysmon events into Elasticsearch and surfaces them through the Hunt interface, enabling correlation across event types using fields such as process.entity_id and winlog.event_data.Image.

Tools and Methods

Tools Used

- Security Onion (Hunt Dashboard): Primary analysis platform used to query, filter, and correlate Sysmon events.
- Sysmon (System Monitor): Windows system service that generated the underlying event data loaded into Security Onion.
- so-import-evtx: Security Onion command-line utility used to import EVTX files into the Elasticsearch backend.
- VirusTotal: Used to contextualize the identified executable (nc.exe / Netcat).
- MITRE ATT&CK Framework: Used to map observed behaviors to known adversary techniques.

Procedure

The Win10Logs.zip archive was extracted, producing a directory named Logs3 containing four EVTX files: Application3.evtx, Security3.evtx, sysmon3_win10.evtx, and System3.evtx. All files were imported into Security Onion from within the Logs3 directory using the following command:

```
sudo so-import-evtx *.evtx
```

After import, the Hunt interface was opened with the time range set to 2021/03/29 - 2021/03/30, as indicated by the import output. For each investigation question, a targeted filter was applied in the Hunt search bar using Onion Query Language (OQL). Events were expanded to reveal full field values and screenshots were captured to document all evidence. Process correlation was performed using winlog.event_data.Image and process.entity_id to link network connections and registry activity back to specific processes.

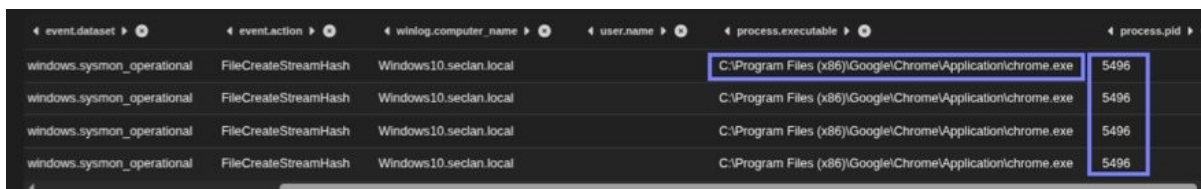
Results

Question 1: Executable Downloaded by Chrome

To identify the executable downloaded by Chrome, the Hunt dashboard was filtered for Sysmon Event ID 15 (file_create_stream_hash). These events are generated when Chrome downloads a file because Chrome automatically attaches a Zone.Identifier Alternate Data Stream to mark all downloaded files as originating from the internet. The filter applied was:

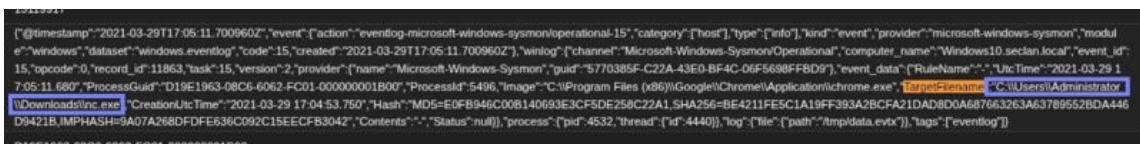
```
winlog.event_id: 15
```

The results returned four matching events, all with process.executable showing C:\Program Files (x86)\Google\Chrome\Application\chrome.exe as the initiating process (Figure 1). Expanding one of these events (Figure 2) revealed the TargetFilename field identifying the downloaded executable as nc.exe, the Windows version of Netcat. Netcat is a well-known networking utility frequently used by adversaries as a backdoor for creating reverse shells and maintaining command-and-control access. The event also recorded the MD5 hash E0FB946C00B140693E3CF5DE258C22A1, a key indicator of compromise.



event.dataset	event.action	winlog.computer_name	user.name	process.executable	process.pid
windows.sysmon_operational	FileCreateStreamHash	Windows10.seclan.local		C:\Program Files (x86)\Google\Chrome\Application\chrome.exe	5496
windows.sysmon_operational	FileCreateStreamHash	Windows10.seclan.local		C:\Program Files (x86)\Google\Chrome\Application\chrome.exe	5496
windows.sysmon_operational	FileCreateStreamHash	Windows10.seclan.local		C:\Program Files (x86)\Google\Chrome\Application\chrome.exe	5496
windows.sysmon_operational	FileCreateStreamHash	Windows10.seclan.local		C:\Program Files (x86)\Google\Chrome\Application\chrome.exe	5496

Figure 1: Sysmon Event ID 15 results confirming Chrome as the process that downloaded nc.exe.



```
[{"timestamp": "2021-03-29T17:05:11.700960Z", "event": {"action": "eventlog-microsoft-windows-sysmon/operational-15", "category": ["host"], "type": ["info"], "kind": "event", "provider": "microsoft-windows-sysmon", "module": "windows", "dataset": "windows.eventlog", "code": 15, "created": "2021-03-29T17:05:11.700960Z", "winlog": {"channel": "Microsoft-Windows-Sysmon/Operational", "computer_name": "Windows10.seclan.local", "event_id": 15, "opcode": 0, "record_id": 11863, "task": 15, "version": 2, "provider": {"name": "Microsoft-Windows-Sysmon", "guid": "5770385F-C22A-43E0-BF4C-06F5698FBD9", "event_data": {"RuleName": "2021-03-29 17:05:11.680", "ProcessGuid": "D19E1963-08C6-6062-FC01-000000001B00", "ProcessId": 5496, "Image": "C:\\Program Files (x86)\\Google\\Chrome\\Application\\chrome.exe", "TargetFilename": "C:\\Users\\Administrator\\Downloads\\nc.exe", "CreationUtcTime": "2021-03-29 17:04:53.750", "Hash": "MD5=E0FB946C00B140693E3CF5DE258C22A1.SHA256=BE4211FE5C1A19FF393A28CFA21DAD8D0A687663263A63789552BDA446D9421B.IMPHASH=9A07A268DFDE636C092C15EECFB3042", "Contents": "", "Status": "null", "process": {"pid": 4532, "thread": {"id": 4440}}, "log": {"file": {"path": "tmp\\data.evtx"}}, "tags": ["eventlog"]}}}, "D19E1963-08C6-6062-FC01-000000001B00"}
```

Figure 2: Expanded Event ID 15 showing nc.exe as the downloaded file, its full path, and MD5 hash.

Question 2: Download Location on the Windows System

The TargetFilename field from the same Event ID 15 record provides the full file system path where the executable was saved. The complete download path observed was:

```
C:\Users\Administrator\Downloads\nc.exe
```

This confirms that nc.exe was saved to the Administrator account's Downloads directory, indicating the Chrome session was running under the Administrator account. Running a browser as Administrator is a poor security practice, as the downloaded file inherits elevated permissions. The evidence is visible in Figure 2 above, where the TargetFilename field displays the full path.

Question 3: First Network Connection from the Downloaded Executable

To trace the first network outreach made by nc.exe, the Hunt dashboard was searched for all events referencing nc.exe, which surfaced the complete event timeline including Sysmon Event ID 3 (network_connection). Figure 3 shows the full nc.exe event list confirming the activity sequence. The filter used to isolate network connection events was:

```
winlog.event_id: 3 AND winlog.event_data.Image: "*nc.exe"
```

The expanded network connection event (Figure 4) revealed the following details:

- Source IP: 192.168.1.3 (Windows10.seclan.local)
- Destination IP: 52.22.39.17
- Destination hostname: ec2-52-22-39-17.compute-1.amazonaws.com
- Destination port: 80 (HTTP)
- Protocol: TCP
- Timestamp: 2021-03-29 02:33:21.738

The destination is an Amazon AWS EC2 instance, a commonly used hosting platform for C2 infrastructure. nc.exe connecting outbound on port 80 to an AWS server is a strong indicator of a reverse shell or C2 callback, as port 80 is frequently used to blend malicious traffic with legitimate HTTP traffic and evade firewall detection.

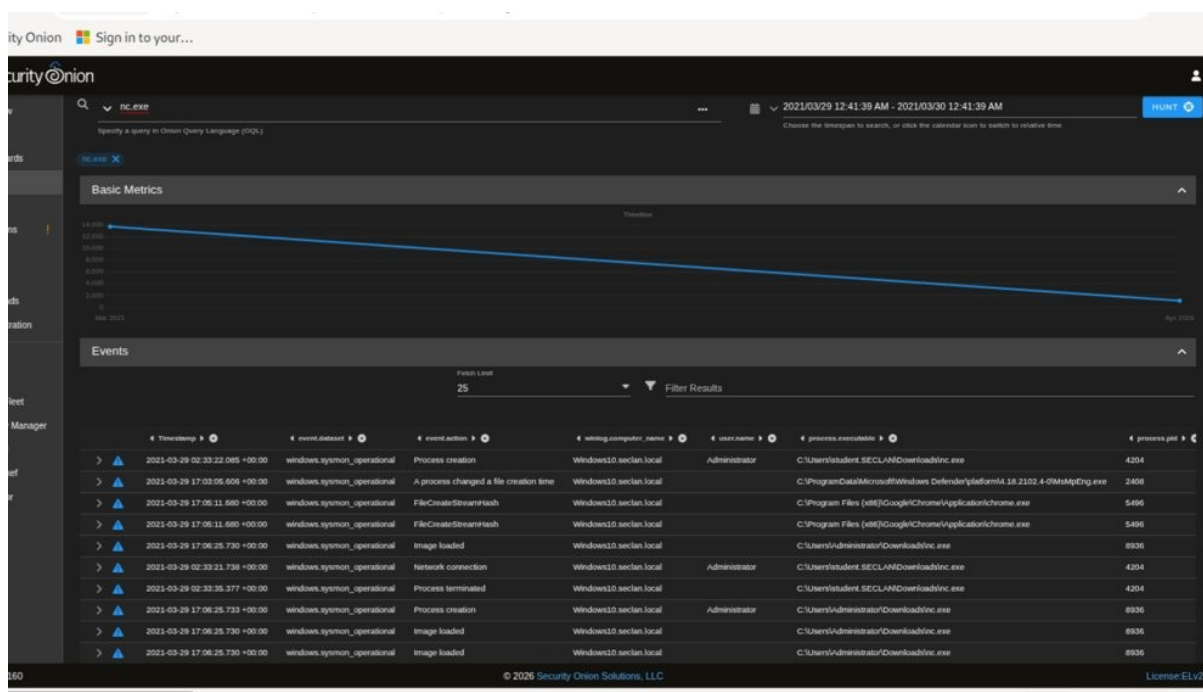


Figure 3: Full nc.exe event list in Security Onion showing process creation, FileCreateStreamHash, and network connection events in chronological order.

destination.domain	ec2-52-22-39-17.compute-1.amazonaws.com
destination.ip	52.22.39.17
destination.port	80

Figure 4: Expanded Sysmon Event ID 3 showing nc.exe initiating an outbound TCP connection to AWS EC2 server 52.22.39.17 on port 80.

Question 4: Process That Changed a File Creation Time

Sysmon Event ID 2 (file_create_time_changed) records instances where a process modifies a file's creation timestamp, a technique known as timestomping. To identify this activity, the following filter was applied:

```
winlog.event_id: 2 AND winlog.event_data.Image: "*nc.exe"
```

The result returned one matching event logged at 2021-03-29 17:03:05.606. Figure 5 shows the expanded event with the responsible process and target file. Figure 6 shows the associated timestamp fields. The key fields observed were:

- Process (Image): C:\ProgramData\Microsoft\Windows Defender\platform\4.18.2102.4-0\MsMpEng.exe
- Target file: C:\Users\student.SECLAN\Downloads\nc.exe
- CreationUtcTime: 2021-03-29 02:32:25.783
- PreviousCreationUtcTime: 2021-03-29 02:32:25.783

The process responsible was MsMpEng.exe, the Windows Defender antimalware service, which triggered this event while scanning nc.exe after it was downloaded. While both timestamps are identical (indicating a benign Defender scan artifact rather than deliberate attacker timestomping), the event is forensically significant as it documents that Windows Defender detected and interacted with nc.exe and confirms the file's presence on disk at that time.

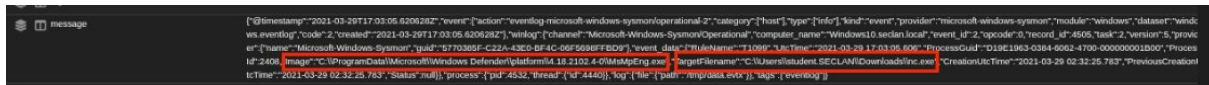


Figure 5: Sysmon Event ID 2 showing MsMpEng.exe as the Image and nc.exe as the TargetFilename.

winlog.computer_name	Windows10.seclan.local
winlog.event_data.CreationUtcTime	2021-03-29 02:32:25.783
winlog.event_data.PreviousCreationUtcTime	2021-03-29 02:32:25.783
winlog.event_id	2
winlog.opcode	0
winlog.provider.guid	5770385F-C22A-43E0-BF4C-06F5698FFBD9
winlog.provider.name	Microsoft-Windows-Sysmon
winlog.record_id	4505
winlog.task	2

Figure 6: Event ID 2 detail showing the CreationUtcTime and PreviousCreationUtcTime fields recorded during the Windows Defender scan of nc.exe.

Question 5: Registry Keys Modified by Explorer.EXE

Registry modifications are a common persistence mechanism. Sysmon Event ID 13 (registry_value_set) records these changes along with the responsible process. To identify registry activity linked to Explorer.EXE, the following filter was applied:

```
winlog.event_id: 13 AND winlog.event_data.Image: "*explorer.exe"
```

The results (Figure 7) confirmed C:\Windows\Explorer.EXE as the modifying process. Expanding that event (Figure 8) revealed the full registry path that was modified:

```
HKUS-1-5-21-832756907-3587282038-2081127716-500\Software\Microsoft\Windows\CurrentVersion\Explorer\FileExts\.exe\OpenWithProgids\exefile
```

This key is located under the user hive and relates to the file extension association for .exe files under the Explorer FileExts path. Modifications here can redirect which program opens .exe files, a technique used for persistence or execution hijacking. The timing of this modification, shortly after nc.exe activity, makes it a relevant finding in the context of this investigation.

```
[{"timestamp": "2021-03-29T17:05:11.701090Z", "event": {"action": "eventlog-microsoft-windows-sysmon/operational-15", "category": "[host]", "type": "[info]", "kind": "event", "provider": "microsoft-windows-sysmon", "module": "windows", "dataset": "windows.eventlog", "code": 15, "created": "2021-03-29T17:05:11.701090Z", "winlog": {"channel": "Microsoft-Windows-Sysmon/Operational", "computer_name": "Windows10.seclan.local", "event_id": 15, "opcode": 0, "record_id": 11866, "task": 15, "version": 2, "provider": {"name": "Microsoft-Windows-Sysmon", "guid": "5770385F-C22A-43E0-BF4C-06F5698FFBD9"}, "event_data": {"RuleName": "", "UtcTime": "2021-03-29 17:05:11.680", "ProcessGuid": "D19E1963-08C6-6062-FC01-000000001B00", "ProcessId": 5496, "ProcessName": "C:\\Program Files (x86)\\Google\\Chrome\\Application\\chrome.exe", "TargetFileName": "C:\\Users\\Administrator\\Downloads\\nc.exe", "CreationUtcTime": "2021-03-29 17:04:53.750", "Hash": "MD5=E0FB946C00B140693E3CF5DE258C22A1, SHA256=BE4211FE5C1A18FF393A2BCFA21DAD8D0A687663263A637895528DA446D9421B, IMPHASH=9A07A268DFDFE636C082C15EECF83042", "Contents": "", "Status": "null"}, "process": {"pid": 4532, "thread": {"id": 4440}}, "log": {"file": {"path": "tmp\\data.evtx"}}, "tags": ["eventlog"]}]
```

Figure 7: Sysmon Event ID 13 results showing C:\Windows\Explorer.EXE as the process that performed a registry value modification.

```
HKUS-1-5-21-832756907-3587282038-2081127716-500\Software\Microsoft\Windows\CurrentVersion\Explorer\FileExts\.exe\OpenWithProgids\exefile
```

Figure 8: Expanded registry event revealing the full HKUS path modified by Explorer.EXE under FileExts for .exe file associations.

Indicators of Compromise (IOC) Summary

Type	Value	Context
File Name	nc.exe	Malicious executable (Netcat) downloaded via Chrome
File Path	C:\Users\Administrator\Downloads\nc.exe	Primary download location (Administrator account)
File Path	C:\Users\student.SECLAN\Downloads\nc.exe	Secondary path detected by Windows Defender
MD5 Hash	E0FB946C00B140693E3CF5DE258C22A1	Hash of nc.exe as recorded in Sysmon Event ID 15
Destination IP	52.22.39.17	AWS EC2 C2 server contacted by nc.exe on port 80
Destination Host	ec2-52-22-39-17.compute-1.amazonaws.com	Resolved hostname of the C2 server
Dest. Port	80 (TCP)	Port used for outbound C2 connection
Registry Key	HKUS...\FileExts\.exe\OpenWithProgids\exefile	Registry value set by Explorer.EXE

Conclusion

The Sysmon log analysis revealed a coherent sequence of adversary activity on the Windows 10 host. Google Chrome was used as the initial delivery vector, downloading nc.exe (Netcat) to the Administrator account's Downloads directory. Netcat is a dual-use networking tool that adversaries commonly use to establish reverse shells and maintain persistent access to compromised systems.

Shortly after the file appeared on disk, nc.exe initiated an outbound TCP connection to an Amazon AWS EC2 instance (52.22.39.17) on port 80, consistent with a C2 callback designed to blend with legitimate HTTP traffic. Windows Defender (MsMpEng.exe) subsequently scanned nc.exe and generated an Event ID 2 record documenting the interaction. A registry modification associated with Explorer.EXE under the FileExts autorun path was also identified, suggesting a persistence mechanism was established to ensure continued access following system reboots.

The observed behaviors map to the following MITRE ATT&CK techniques:

- T1204.002 (User Execution: Malicious File): nc.exe delivered and executed via browser download.
- T1059 (Command and Scripting Interpreter): Netcat used to establish an interactive shell session.
- T1071.001 (Application Layer Protocol: Web Protocols): C2 communication carried over port 80 HTTP.
- T1070.006 (Indicator Removal: Timestamp): File creation time modification recorded on nc.exe.
- T1547.001 (Boot/Logon Autostart: Registry Run Keys): Registry key modified for persistence via Explorer.EXE.

This investigation demonstrates the critical forensic value of deploying Sysmon on endpoint hosts. Without Sysmon, correlating a downloaded file with a specific outbound network connection and subsequent registry change would not be achievable using standard Windows Event Logging alone.

Non-Technical Summary

A malicious program called nc.exe was secretly downloaded onto a Windows computer through the Google Chrome web browser. Once on the computer, this program reached out over the internet to a remote server hosted on Amazon's cloud platform, a common technique attackers use to remotely control a compromised computer. The program also triggered changes to the Windows registry, a database that controls how the computer behaves, which attackers use to keep their program running even after the computer is restarted. Windows Defender detected the file during a routine scan. This type of attack highlights the importance of monitoring files

downloaded through web browsers and watching for unusual programs making outbound connections to external servers.

References

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